

A Block-Partition Criterion For Weighted Lower Semi-Frames

Automatic math-discovery smoke test

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Source Problem

The source paper is E. Zikkos, *The Gaussian Gabor system at the critical density is a weighted lower semi frame*, arXiv:2606.00764, Functional Analysis.

It quotes Conjecture 3.1 of Balazs–Corso–Stoeva, *Weighted frames, weighted lower semi frames and unconditionally convergent multipliers*, arXiv:2310.18957:

Theorem 1.2 answers partially the following conjecture.

Conjecture 1.1. *[1, Conjecture 3.1] If a family $\mathcal{F}$ in a separable Hilbert space $\mathcal{H}$ is complete, then $\mathcal{F}$ is a weighted lower semi frame for $\mathcal{H}$.*

The conjecture is known to hold if $\mathcal{F}$ has a reproducing partner for $\mathcal{H}$ ([1, Proposition 3.3]). A special case is when $\mathcal{F}$ contains a Schauder basis for $\mathcal{H}$ or a frame for $\mathcal{H}$ ([1, Corollary 3.5]). Corollary 1.2 adds a further case: any complete family containing a complete and minimal subfamily.

Remark 1.4. *The case when $\mathcal{F}$ is complete but every complete subfamily has infinite excess remains open.*

A natural example is the Müntz family $\{t^{\lambda_n}\}$, where $\{\lambda_n\} \subset \mathbb{N}$ and $\sum \lambda_n^{-1} = \infty$. By the Müntz-Szász theorem the family is complete in $L^2(0, 1)$ if and only if $\sum \lambda_n^{-1} = \infty$. Thus, every complete subfamily of $\{t^{\lambda_n}\}$ again has an infinite excess in $L^2(0, 1)$. Whether such families are weighted lower semi frames for $L^2(0, 1)$ is unknown.

Thus the ambient problem is whether every complete family in a separable Hilbert space is a weighted lower semi-frame. The same passage notes that the complete-minimal case is known and that the infinite-excess case remains open; it also highlights the Müntz family $\{t^{\lambda_n}\} \subset L^2(0, 1)$, with $\sum_n \lambda_n^{-1} = \infty$, as a natural test case.

Result Type

This packet is a **partial result**. It gives a full affirmative answer for the Müntz-family example in the quoted passage, and it proves a general block-partition criterion, but it does not prove the full complete-family conjecture.

Definitions And Convention

A sequence (g_n) in $\mathcal{H}$ is a lower semi-frame with lower bound $A > 0$ if

$$A \|f\|^2 \leq \sum_n |\langle f, g_n \rangle|^2 \quad (f \in \mathcal{H}).$$

The sum is an extended nonnegative series. In particular, if the sum is $+\infty$ for every nonzero f , then the lower semi-frame inequality holds for every finite $A > 0$. A sequence (f_n) is a weighted lower semi-frame if $(\omega_n f_n)$ is a lower semi-frame for some positive weights ω_n .

Block-Partition Criterion

Theorem 1. *Let $(f_n)_{n \in \mathbb{N}}$ be a sequence in a separable Hilbert space $\mathcal{H}$. Suppose that*

$$\mathbb{N} = \bigsqcup_{j=1}^{\infty} I_j$$

and that each subfamily $(f_n)_{n \in I_j}$ is complete in $\mathcal{H}$. Then (f_n) is a weighted lower semi-frame. More precisely, for every $A > 0$ there are positive weights (ω_n) such that

$$\sum_n |\langle f, \omega_n f_n \rangle|^2 = +\infty$$

for every nonzero $f \in \mathcal{H}$.

Proof. Fix an orthonormal basis (e_m) of $\mathcal{H}$. Since the j -th block is complete, choose a finite subset $G_j \subset I_j$ such that

$$\text{dist}(e_m, \text{span}\{f_n : n \in G_j\}) < \frac{1}{j} \quad (1 \leq m \leq j).$$

Let

$$E_j = \text{span}\{f_n : n \in G_j\}$$

and let P_j be the orthogonal projection onto E_j . The preceding approximation property implies $P_j \rightarrow I$ strongly: first check this on finite linear combinations of basis vectors, then use uniform boundedness of the projections and density.

For each j , the finite family $(f_n)_{n \in G_j}$ spans E_j . Therefore

$$\delta_j = \inf_{\substack{x \in E_j \\ \|x\|=1}} \sum_{n \in G_j} |\langle x, f_n \rangle|^2$$

is strictly positive. Define

$$\omega_n = \left(\frac{A}{\delta_j} \right)^{1/2} \quad (n \in G_j),$$

and assign any positive value, say 1, to indices not belonging to $\bigcup_j G_j$. The sets G_j are pairwise disjoint because the blocks I_j are disjoint.

For every $f \in \mathcal{H}$,

$$\sum_{n \in G_j} |\langle f, \omega_n f_n \rangle|^2 = \frac{A}{\delta_j} \sum_{n \in G_j} |\langle P_j f, f_n \rangle|^2 \geq A \|P_j f\|^2.$$

If $f \neq 0$, then $\|P_j f\| \rightarrow \|f\| > 0$. Hence the block contributions are bounded below by a positive constant for all large j , and summing over the disjoint blocks gives

$$\sum_n |\langle f, \omega_n f_n \rangle|^2 = +\infty.$$

□

Application To The Muntz Family

Lemma 1. *Let (a_n) be a positive sequence with $\sum_n a_n = \infty$. Then $\mathbb{N}$ can be partitioned into countably many disjoint sets J_k such that $\sum_{n \in J_k} a_n = \infty$ for every k .*

Proof. Split $\mathbb{N}$ into consecutive finite intervals B_m such that $\sum_{n \in B_m} a_n \geq 1$ for every m . This is possible because every tail of the positive divergent series diverges. Choose a map $s : \mathbb{N} \rightarrow \mathbb{N}$ such that $s^{-1}(k)$ is infinite for every k , and set

$$J_k = \bigcup_{m:s(m)=k} B_m.$$

These sets are disjoint, cover $\mathbb{N}$, and each corresponding subseries dominates an infinite sum of ones. $\square$

Proposition 1. *Assume $\lambda_n \in \mathbb{N}$ and $\sum_n \lambda_n^{-1} = \infty$. Under the Muntz–Szász completeness criterion used in arXiv:2606.00764, the family*

$$\{t^{\lambda_n}\} \subset L^2(0, 1)$$

is a weighted lower semi-frame for $L^2(0, 1)$.

Proof. Apply the lemma to $a_n = \lambda_n^{-1}$. The resulting partition $\mathbb{N} = \bigsqcup_k J_k$ satisfies

$$\sum_{n \in J_k} \lambda_n^{-1} = \infty \quad (k \geq 1).$$

By the Muntz–Szász criterion as invoked in the source paper, each subfamily $\{t^{\lambda_n} : n \in J_k\}$ is complete in $L^2(0, 1)$. The block-partition theorem now gives positive weights making the full Muntz family a lower semi-frame. $\square$

Why This Is Not A Full Solution

The argument requires a partition of the index set into countably many complete subfamilies. That condition holds for the Muntz example because the divergent series $\sum_n \lambda_n^{-1}$ can be split into countably many divergent subseries. It is not shown here, nor in the source paper, that every complete family in the remaining infinite-excess regime admits such a partition.

Thus the result should remain in the `partial` directory. It would become a full solution of the complete-family conjecture only after removing the block-partition hypothesis or proving that hypothesis for all complete families covered by the conjecture.

Human Review Recommendation

The proof is short and appears internally consistent. The main points to check are the lower semi-frame convention in the two cited papers, the finite-dimensional lower bound $\delta_j > 0$, and the exact Muntz–Szász completeness statement used for subfamilies of $\{t^{\lambda_n}\}$.